

29 For diffraction of the electron beam, the de Broglie wavelength is $\lambda = \frac{h}{p} \approx 10^{-10} \text{ m}$.

$$p = \frac{h}{10^{-10}}$$

$$(9.11 \times 10^{-31})v = \frac{6.63 \times 10^{-34}}{10^{-10}}$$

$$v = 7.28 \times 10^6 \text{ m s}^{-1}$$

Therefore the order of magnitude of velocity required is 10^6 m s^{-1} .

Answer: C